1.Spring Data JPA - Quick Example

```java
import java.util.List;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
import com.cognizant.orm_learn.model.Country;
import com.cognizant.orm_learn.service.CountryService;

@SpringBootApplication
public class OrmLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(OrmLearnApplication.class);

    private static CountryService countryService;

    public static void main(String[] args) {
        ApplicationContext context =
            SpringApplication.run(OrmLearnApplication.class, args);
        LOGGER.info("Inside main");
        countryService = context.getBean(CountryService.class);
        testGetAllCountries();
    }

    private static void testGetAllCountries() {
        LOGGER.info("Start");
        List<Country> countries = countryService.getAllCountries();
        LOGGER.debug("countries={}", countries);
        LOGGER.info("End");
    }
}
```

orm-learn > src > main > java > com > cognizant > orm_learn > repository > CountryRepository.java

```java
package com.cognizant.orm_learn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;
import com.cognizant.orm_learn.model.Country;

@Repository
public interface CountryRepository extends JpaRepository<Country, String> {

}
```

application.properties | Country.java | CountryRepository.java | CountryService.

orm-learn > src > main > java > com > cognizant > orm_learn > model > Country.java

```java
package com.cognizant.orm_learn.model;

import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name = "country")
public class Country {

    @Id
    @Column(name = "co_code")
    private String code;

    @Column(name = "co_name")
    private String name;

    public String getCode() { return code; }
    public String getName() { return name; }
    public void setCode(String code) { this.code = code; }
    public void setName(String name) { this.name = name; }

    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}
```

```
01-07-26 13:15:16.279 restartedMain        INFO c.c.o.OrmLearnApplication       test
01-07-26 13:15:16.369 restartedMain       DEBUG org.hibernate.SQL
0
01-07-26 13:15:16.394 restartedMain       DEBUG c.c.o.OrmLearnApplication       test
 [code=UK, name=United Kingdom], Country [code=US, name=United States of America]]
01-07-26 13:15:16.396 restartedMain        INFO c.c.o.OrmLearnApplication       test
01-07-26 13:15:16.399 licationShutdownHook  INFO rEntityManagerFactoryBean
unit 'default'
01-07-26 13:15:16.401 licationShutdownHook  INFO c.z.h.HikariDataSource
01-07-26 13:15:16.406 licationShutdownHook  INFO c.z.h.HikariDataSource
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.455 s
[INFO] Finished at: 2026-07-01T13:15:16+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\SARO\Downloads\ORM-LEARN\orm-learn>
```

```
    ->     co_code VARCHAR(2) PRIMARY KEY,
    ->     co_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO country VALUES ('IN', 'India');
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO country VALUES ('US', 'United States of America');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO country VALUES ('UK', 'United Kingdom');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM country;
+---------+--------------------------+
| co_code | co_name                  |
+---------+--------------------------+
| IN      | India                    |
| UK      | United Kingdom           |
| US      | United States of America |
+---------+--------------------------+
3 rows in set (0.00 sec)

mysql>
```

2.Difference between JPA, Hibernate and Spring Data JPA

JPA

1. It is a specification (set of rules) for ORM.
2. It defines how Java objects are mapped to database tables.
3. It cannot work on its own.
4. It uses EntityManager for database operations.
5. Example: @Entity, @Id, EntityManager.

Hibernate

1. It is an ORM framework.
2. It is an implementation of JPA.
3. It converts Java objects into SQL queries.
4. It can work without Spring.
5. It provides features like HQL, caching, and lazy loading.

Spring Data JPA

1. It is a Spring module built on top of JPA.
2. It reduces boilerplate code for database operations.
3. It requires a JPA implementation such as Hibernate.
4. It uses JpaRepository for CRUD operations.
5. It automatically provides methods like save(), findById(), and findAll().